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Manufacturing ERP Implementation Specialist

Discovery-Led Odoo MRP for Multi-Stage Fabrication

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Samundri, Punjab, Pakistan | Remote Worldwide

Prepared for

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Executive Summary

This document presents my credentials, relevant experience, and proposed approach for your steel container manufacturing Odoo ERP engagement - structured discovery first, configuration second.

Your operation spans multiple interconnected production stages - material preparation, machining, welding, painting, and assembly - serving waste, construction, and municipal customers with roll-off containers, dumpsters, and custom builds. Replacing manual and spreadsheet-based tracking with Odoo requires more than module installation: it requires accurately translating real shop-floor operations into work centers, routings, BOMs, work orders, and quality checkpoints that your production team will actually use.

I specialize in exactly this type of engagement. My approach begins with structured discovery interviews alongside supervisors, machine leads, and administrative staff - documenting workstations, shifts, order flow, and stage-by-stage handoffs before any Odoo configuration begins.

Alignment With Your Project Structure

- Phase 1 - Discovery: Structured interviews, process mapping, and a documented Odoo module/data plan validated against your internal discovery framework.
- Phase 2 - Configuration & Rollout: MRP, Inventory, Purchase, Quality, Maintenance, Timesheets, and Barcode setup; data migration; testing; training; and go-live support with realistic adoption planning.

Metric	Value
Odoo Implementations	91+
Years Experience	6+
Global Clients	70+
Client Rating	5.0 / 5.0

Odoo Certified Developer | Upwork Top Rated | 100% Job Success | MSSE @ Quantic, USA

Why This Engagement Fits

Direct mapping of your stated requirements to demonstrated capabilities and deliverables.

Your Requirement	My Evidence & Approach
Structured discovery with production supervisors	4-week discovery framework; PM experience at Cavi Consulting; personally lead all stakeholder interviews
Work centers, routings, BOMs, work orders	Partsnord Europe AB (automotive parts); MRP II solution patterns; Product Configurator with dynamic BOM generation
Quality checkpoints at production stages	Odoo QMS, LIMS, and Batch Manufacturing & Traceability module expertise
Maintenance module for equipment	CMMS, Predictive Maintenance, and Equipment Rental maintenance tracking implementations
Barcode scanning on shop floor	Warehouse Automation Suite; Inventory Mastery Suite (13 custom inventory modules)
Flexible material consumption (leftover sheet metal reuse)	Batch Production, Multi-Warehouse Replenishment, scrap/leftover location rules, alternative component handling
KPI dashboards and reporting	Plectar engagement: 413+ hours building Odoo Spreadsheet reports; cycle time, utilization, and delivery KPIs
Production status pipeline mirroring real stages	Car Repair Workshop System - multi-stage job tracking pipeline; direct parallel to your Pending → Shipped workflow
Phase 2: second business unit (servicing/upfits)	Multi-Company Consolidation; Odoo_Base_Multi_Company module; parallel routing templates
Odoo Studio custom fields (gauge, dimensions, weight)	90+ Odoo custom module builds; OWL, XML, Python - minimal customization principle

Implementation Methodology

Discovery Interviews → Process Map & Odoo Data Plan → MRP, Inventory, Purchase, Quality → Pilot & UAT → Training & Go-Live → Post-Live Optimization

Professional Profile

I am a Senior Odoo Developer and ERP Implementation Specialist with 6+ years of experience delivering production-grade Odoo solutions for clients across 25+ countries. As CEO and Founder of CODCrafters, I lead a team that ships enterprise ERP transformations, custom module development, and business process automation.

Odoo is my core specialty - 91+ dedicated Odoo projects across versions 15 through 19, spanning Manufacturing (MRP), Inventory, Purchase, Quality, Maintenance, Timesheets, Barcode, Accounting, and custom reporting via Odoo Spreadsheets.

Metric	Count
Total Projects	185+
Odoo Builds	91+
GitHub Repos	192
Countries Served	25+

Credential	Details
Odoo Certified Developer	Official Odoo certification - custom modules, enterprise implementations, OWL framework
Upwork Top Rated	100% Job Success Score, 5.0/5.0 overall rating across 70+ clients
MSSE - Quantic School of Business & Technology	Master of Software Systems Engineering, Washington D.C., USA (in progress)
GitHub - 192 Public Repositories	github.com/UsmanGhias - Odoo modules, integrations, and open-source contributions

Core Competencies

- Odoo Manufacturing (MRP) - work centers, routings, BOMs, work orders
- Inventory & warehouse management with barcode operations
- Quality management - inspection points, NC tracking, CAPA workflows
- Maintenance (CMMS) - preventive schedules, downtime tracking
- Business process discovery and documentation
- Odoo Spreadsheets - dynamic KPI dashboards and reporting
- Python, PostgreSQL, XML, OWL, JavaScript, REST/XML-RPC APIs
- ERP integrations - QuickBooks, DocuSign, n8n automation

Current & Recent Engagements

Named client references demonstrating active, relevant Odoo manufacturing and reporting experience.

Partsnord Europe AB - Senior Odoo Developer

October 2025 – June 2026 | Europe (Remote) | Full-time

Senior Odoo Developer specializing in enterprise-level implementations and custom module development for the automotive parts industry.

- Leading enterprise-level Odoo implementations for discrete manufacturing environments
- Developing custom Odoo modules tailored to automotive parts production workflows
- Managing technical architecture and coordinating development teams

Relevance: *Discrete manufacturing, parts inventory, production-adjacent ERP - directly applicable to your multi-stage fabrication environment.*

Plectar - Odoo Spreadsheet Reporting Expert

December 2025 – Present | Remote | 530+ hours delivered at $24/hr

Long-running engagement building dynamic, data-driven reporting on top of the Odoo Spreadsheet module.

- Designed advanced data models and complex logic for reusable KPI reports
- Built management dashboards for operational decision-making inside Odoo
- Consistently rated 5.0/5.0 for technical proficiency, communication, and efficiency

Relevance: *Direct match to your KPI dashboard requirements - on-time delivery, cycle time, utilization, and material usage reporting.*

Hunter - Odoo Expert

December 2025 – Present | Remote | Contract

Providing expert Odoo consultation and development services. Specializing in ERP customization, business process optimization, and system integration.

Relevance: *ERP customization and process optimization across varied manufacturing and operations contexts.*

Cavi Consulting - Odoo Project Manager & Developer

August 2025 – Present | Austria (Remote) | Part-time

Senior Odoo Project Manager and Developer managing enterprise Odoo implementations with AI integration.

- Managing enterprise Odoo implementations with AI integration
- Leading development teams in AI-powered Odoo solutions
- Coordinating project delivery, stakeholder communication, and rollout planning

Relevance: *Project management and enterprise rollout experience - critical for your phased discovery-to-go-live engagement.*

Sisters Code - Odoo Web Development

October 2025 | United States (Remote) | Freelance

Complete Odoo implementation from start to finish with custom Odoo web modules.

- End-to-end Odoo implementation - discovery through go-live
- Custom Odoo web module development using OWL framework
- Full stakeholder engagement and training delivery

Relevance: *Demonstrates ability to own full implementation lifecycle independently.*

Engagement Model

- Solo lead: I personally conduct all discovery interviews and lead configuration.
- CODCrafters bench: Additional developers available for parallel configuration or data migration if scope expands.

Manufacturing-Relevant Case Studies

Selected engagements demonstrating discrete manufacturing, job-shop tracking, and operational reporting capabilities.

Automotive Parts - Discrete Manufacturing

[Partsnord Europe AB]

Challenge

A European automotive parts manufacturer needed to coordinate multi-SKU parts production, maintain inventory accuracy across warehouses, and replace fragmented manual tracking with a centralized ERP system.

Solution

Led Odoo MRP-adjacent configuration including work center setup, production routings, purchase and inventory integration, and custom module development tailored to automotive parts workflows.

Outcome

Centralized production visibility across the operation. Reduced reliance on manual tracking and spreadsheet-based coordination. Established a scalable Odoo architecture supporting ongoing module expansion.

Car Repair Workshop System

[Automotive · Job-Shop]

Challenge

An automotive repair chain was losing revenue to manual job card tracking, incorrect billing, and slow customer communication. No real-time visibility into job status across multiple service bays.

Solution

Delivered a custom Odoo workshop module with real-time job card tracking, multi-stage status pipeline, automated invoicing, and customer SMS notifications.

Outcome

Significantly faster invoice processing with near-elimination of manual billing errors. Direct parallel to your production pipeline: Pending → stage transitions → Ready to Ship → Shipped.

Odoo Spreadsheet KPI Reporting

[Plectar]

Challenge

Management relied on reporting systems outside Odoo, resulting in inconsistent KPIs, delayed decision-making, and duplicated data entry.

Solution

Built dynamic Odoo Spreadsheet reports with advanced data models. Migrated the client's entire internal reporting infrastructure into Odoo.

Outcome

Management dashboards now live inside Odoo with real-time data. Client consistently rates engagement 5.0/5.0 across 413+ delivered hours.

Supporting Module Expertise

Module / Solution	Capability
Material Requirements Planning (MRP II)	Capacity planning, master scheduling, demand forecasting
Warehouse Automation Suite	Barcode scanning, automated picking, smart routing
Quality Management System (QMS)	ISO compliance, non-conformance handling, CAPA workflows
Maintenance Management (CMMS)	Preventive maintenance, spare parts tracking, downtime analysis
Batch Manufacturing & Traceability	Batch genealogy, quality hold management, scrap tracking
Product Configurator	Dynamic BOM generation from product options
Inventory Mastery Suite	13 custom modules: real-time tracking, warehouse control
Job Costing Management	Make-to-order job costing, material and labor allocation
Material Purchase Requisition	Material requisition workflows from shop floor to purchasing

Discovery Methodology

My approach to running discovery with non-technical shop floor staff - structured, respectful of operational expertise, and designed to produce actionable Odoo configuration blueprints.

Week	Activities	Stakeholders	Deliverable
1	Facility walkthrough, order-to-ship shadowing	Production supervisors, shift leads	Current-state process swimlanes
2	Workstation interviews - one session per production stage	Machine leads, welders, paint crew, assembly	Work center map with capacity per station
3	BOM and material deep dive; purchasing workflows; QC points	Admin staff, purchasing, QC team	BOM structure + consumption rules (incl. leftover sheet reuse)
4	Reporting requirements workshop; adoption planning; pilot selection	Management, floor representatives	Odoo data model blueprint + phased rollout plan

Interview Principles

- Shop-floor language first - document how supervisors describe their work; translate to Odoo afterward
- Photo and sketch workstations during walkthroughs to anchor work center definitions
- Validate with a pilot product line before configuring all routings and BOMs
- Document exceptions explicitly - leftover sheet metal reuse, rush orders, rework loops
- Sign-off gate - no MRP configuration begins until discovery documentation is approved
- Respect existing workflows - gradual adoption rather than overnight process overhaul

Your Production Stage Mapping

Pending → Preparing Material → Machineshop → Building → Prep & Coating → Ready to Ship → Shipped

+ Accounting status: Open Balance / Released | + Delivery confirmation status

Odoo mapping: Each stage becomes a routing operation on the manufacturing order, with assigned work centers, expected duration, quality checkpoints, barcode scan events, and labor time recording via timesheets.

Proposed Technical Approach

Configuration plan tailored to steel container manufacturing - roll-off containers, dumpsters, and custom builds.

Manufacturing (MRP)

- Work centers defined per real station - material prep, CNC/machining, welding bays, paint booth, assembly
- Routings per product family: standard roll-off, dumpster, and custom build
- BOMs structured by material type: steel sheets (with gauge tracking), structural parts, purchased hardware, consumables
- Flexible consumption rules for leftover sheet metal - scrap/leftover locations, alternative component rules
- Capacity planning per work center with shift-aware scheduling

Inventory & Warehouse

- Multi-location warehouse: raw steel, WIP per stage, finished goods, scrap/leftover inventory
- Barcode operations for material issue, WO start/stop, inter-stage transfers, shipping
- Lot tracking for steel batches and traceability through production stages
- Automated replenishment rules for consumables and purchased hardware

Quality, Maintenance & Labor

- Quality checkpoints at weld inspection, coating test, and final assembly - with NC tracking
- Maintenance module - equipment registry tied to work centers, PM schedules, downtime recording
- Timesheets - labor cost allocation per work order and work center

Custom Fields (Odoo Studio)

- Steel gauge, dimensions (L×W×H), weight per job - via Studio with minimal customization
- Use standard Odoo fields wherever possible; Studio only for industry-specific attributes

Reporting & Dashboards

- On-time delivery performance
- Cycle time per production stage
- Work center utilization rates
- Scrap and defect rates by stage
- Labor efficiency per work order
- Material usage vs. BOM variance

Phase 2 - Second Business Unit

- Parallel routing template for equipment servicing, upfits, and warranty work
- Diagnostic fee tracking and warranty coverage rules
- Approval workflows for warranty claims and service authorizations
- Multi-company structure for operational separation with consolidated reporting

Rollout & Change Management

- Pilot first: One product line, one shift, full workflow before expanding
- Champions per stage: One supervisor per production stage as Odoo advocate
- Barcode gradually: Start with shipping and material issue; expand to all stage transfers
- Parallel run: Keep existing tracking alongside Odoo for 2–4 weeks during pilot
- Weekly floor check-ins: First month post-go-live to resolve friction points

Module & Technology Expertise

Odoo Module	Scope
Manufacturing (MRP)	Work orders, routings, BOMs, capacity planning
Inventory	Multi-location, lot tracking, replenishment
Purchase	RFQs, vendor management, procurement workflows
Quality	Inspection points, checklists, NC tracking
Maintenance	Equipment records, PM schedules, downtime
Timesheets	Labor costing, WO time recording
Barcode	Floor scanning, material movement, WO tracking
Accounting	Integration with production costing and invoicing
Odoo Studio	Custom fields without over-engineering

Category	Technologies
Languages	Python, JavaScript, TypeScript, XML, SQL
Frameworks	Odoo OWL, QWeb, PostgreSQL, Django REST
APIs	REST, XML-RPC, GraphQL, Odoo JSON-RPC
Odoo Versions	15, 16, 17, 18, 19 - Community and Enterprise
DevOps	Docker, CI/CD, AWS, Odoo.sh deployment

Selected GitHub Repositories

- job_costing_managment - job costing for make-to-order
- Material_Purchase_Requistion - material requisition workflows
- Multi_Location_Odoo_17 - multi-location inventory
- Odoo_Base_Multi_Company - multi-entity setups
- cc_kpi - KPI reporting module
- custom-invoice-report - operational reporting
- odoo-n8n - Odoo automation integration

Engagement Model & Commercials

Item	Details
Discovery Lead	Muhammad Usman Khan - personally conducts all supervisor interviews
Team Capacity	CODCrafters bench available for parallel work if needed
Discovery Phase Rate	$22/hour (reduced rate for initial phase)
Standard Rate	$24/hour
Availability	Immediate - remote worldwide, flexible timezone overlap
Communication	Weekly demos, milestone updates, staging access, progress reports

Phase	Scope	Deliverable
Phase 1 - Discovery	4-week interviews, process mapping, BOM analysis	Documented process map and Odoo data plan
Phase 2 - Configuration	MRP, Inventory, Purchase, Quality, Maintenance, Barcode	Configured Odoo with pilot product line
Phase 3 - Rollout	Migration, UAT, training, go-live support	Live system with trained team and documentation

Client Testimonial

"*Working with Usman and the CODCrafters team was seamless from start to finish. Their Odoo expertise transformed our operations, and the custom modules they built were exactly what we needed. Professional, reliable, and highly skilled.*"

- Marlo Renchor, Business Owner, USA | 6-month engagement | 100% Job Success

Recommended Next Steps

1. Share your internal discovery framework - I will review and align my 4-week discovery plan within 48 hours
2. Confirm engagement terms - budget model (hourly vs. fixed), hours per week, target go-live date
3. Schedule Week 1 activities - facility walkthrough and initial supervisor interviews
4. Identify pilot product line - select one product family for first routing, BOM, and work center configuration
5. Confirm Odoo environment - edition, hosting, and existing data sources for migration

Resources & References

- Portfolio: https://usman.codcrafters.org
- GitHub: https://github.com/UsmanGhias (192 public repositories)
- Problem Solution Document: https://docs.google.com/document/d/1TJdZYc7oMWCBv7C6QjMgILOPizTDw1OA2e5i_LCvS4M/edit
- Video Overview: https://www.loom.com/share/f1858a28f08843d9aa7da86c82bc2721

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